

THE MAGI ROOT OBJECT:

Hybrid cDDM with Autonomous Mutant Manifold

The MAGI Meta-Optimization Engine

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Abstract

This document formally defines the mathematical architecture of the Cortico-Cerebellar Evidence Accumulation system, derived via the MAGI protocol. By hybridizing a deterministic Drift-Diffusion Model (cDDM) with an autonomously fluctuating, Golgi-gated stochastic manifold, we formally resolve the topological paradox of high-frequency precision and macroscopic fatigue. The breakthrough of this formulation is the continuous convolutional generative evaluation, which analytically prevents the log-likelihood catastrophe caused by trial-by-trial physiological variance.

1 I. The Cortical Value Basis (Deterministic Intent)

The neocortex and striatum compute a continuous, deterministic low-dimensional value basis. This value generates the Reward Prediction Error ($\delta^{(k)}$) and drives the primary directional Drift Rate ($v^{(k)}$) of the cognitive decision process for trial k .

$$\delta^{(k)} = \text{Reward}^{(k)} - Q_{ctx}^{(k)} \quad (1)$$

$$v^{(k)} = \kappa \cdot Q_{ctx}^{(k)} \quad (2)$$

Decision Time Density: The cognitive accumulation resolves over time τ according to the standard Wiener First Passage Time (WFPT). It is anchored with zero intrinsic non-decision time, as motor latency is handled separately:

$$f_{DT}(\tau) = f_{WFPT}(\tau \mid v^{(k)}, a, w) \quad (3)$$

2 II. The Cerebellar Variance Generator (Stochastic Expansion)

The granular layer acts as an autonomous, uncoupled reservoir generating physiological variance.

Intrinsic Drive: N uncoupled dimensions of continuous Poisson shot noise with basal rates λ :

$$\zeta^{(k)} \sim \text{PoissonShotNoise}(\lambda) \quad (4)$$

Sigmoidal Golgi Gating: The instantaneous demand for computational complexity ($\mathcal{I}^{(k)}$) is defined by recent cortical volatility. This drives the global inhibitory conductance ($\theta^{(k)}$) via a non-linear sigmoid, protecting the system from overfitting pure noise during stable periods:

$$\mathcal{I}^{(k)} = |\delta^{(k-1)}| \quad (5)$$

$$\theta^{(k)} = \frac{\theta_{max}}{1.0 + \exp(-\beta_g \cdot (\mathcal{I}^{(k)} - \mathcal{I}_{bias}))} \quad (6)$$

The Uncoupled Mutant Manifold: The stochastic drive is unmasked against the Golgi threshold and smeared across time via the log-uniform temporal leak matrix $\mathbf{\Lambda}$:

$$\tilde{\mathbf{h}}^{(k)} = \text{ReLU} \left(\zeta^{(k)} - \theta^{(k)} \mathbf{1} \right) \quad (7)$$

$$\mathbf{h}^{(k)} = (\mathbf{I} - \mathbf{\Lambda})\mathbf{h}^{(k-1)} + \mathbf{\Lambda}\tilde{\mathbf{h}}^{(k)} \quad (8)$$

3 III. The Unified Lasso Readout (Motor Automation)

The unified Purkinje and Deep Cerebellar Nuclei (DCN) readout projects the active high-dimensional manifold ($V_{cb}^{(k)} = \mathbf{W}^{(k)}\mathbf{h}^{(k)}$) to dynamically govern the motor execution parameters.

Mean Execution Latency (μ_{tnd}): Disinhibition of the DCN collapses the motor delay from a slow cortical default:

$$\mu_{tnd}^{(k)} = t_{nd_max} - \gamma_{cb} \cdot |V_{cb}^{(k)}| \quad (9)$$

Execution Variance (σ_{tnd}): The physiological jitter is strictly proportional to the total unmasked energy (the $\mathcal{L}_1$ norm) of the active granular manifold:

$$\sigma_{tnd}^{(k)} = \eta_{var} \cdot \|\mathbf{h}^{(k)}\|_1 \quad (10)$$

Motor Time Density: The physical execution time t forms a normal probability distribution:

$$f_{MT}(t) = \mathcal{N} \left(t \mid \mu_{tnd}^{(k)}, \sigma_{tnd}^{(k)} \right) \quad (11)$$

4 IV. The Plasticity Operator (Heterosynaptic Subgradient Descent)

The Purkinje readout weights update online upon receipt of the Climbing Fiber signal (the cortical RPE). The objective minimizes error while ruthlessly pruning unused connections via the $\mathcal{L}_1$ Lasso penalty (λ_{lasso}):

$$\Delta \mathbf{W}^{(k)} = \alpha \cdot \delta^{(k)} \cdot \left(\mathbf{h}^{(k)} \right)^\top - \eta_{decay} \cdot \lambda_{lasso} \cdot \text{sgn} \left(\mathbf{W}^{(k)} \right) \quad (12)$$

$$Q_{ctx}^{(k+1)} = Q_{ctx}^{(k)} + \alpha_{ctx} \cdot \delta^{(k)} \quad (13)$$

5 V. The Convoluted Generative Evaluation (The Topological Filter)

To evaluate the true likelihood of the observed Reaction Time ($RT^{(k)}$) without triggering the $-\infty$ density cliff of trial-by-trial jitter, the model integrates the product of the deterministic decision time and the stochastic motor time via continuous convolution:

$$P(RT^{(k)}) = (f_{DT} * f_{MT})(RT^{(k)}) \quad (14)$$

$$P(RT^{(k)}) = \int_0^{RT^{(k)}} f_{WFPT}(\tau \mid v^{(k)}, a, w) \cdot \mathcal{N} \left(RT^{(k)} - \tau \mid \mu_{tnd}^{(k)}, \sigma_{tnd}^{(k)} \right) d\tau \quad (15)$$

The global objective function for the Neural Architecture Search (NAS) is the minimization of the Negative Log-Likelihood over all K trials:

$$\mathcal{L}_{MAGI} = - \sum_{k=1}^K \log P(RT^{(k)}) \quad (16)$$

System Validation Summary:

Gate 0 (Rademacher): Sustained by the Sigmoidal Golgi Gate ($\theta^{(k)}$).

Gate 2 (Macroscopic Fatigue): Sustained by the Uncoupled Leak Matrix ($\mathbf{\Lambda}$) driving the μ_{tnd} collapse.

Gate 1 (Information Density): Unlocked by the Convolutional Filter ($f_{DT} * f_{MT}$), absorbing the physiological manifold variance (σ_{tnd}) as valid probability mass.